

CLUSTERING

CLUSTER VALIDITY – PART I

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CLUSTER VALIDITY

The following concepts will be introduced:

- ✓ **CLUSTER VALIDATION OR CLUSTER VALIDITY**
- ✓ **TYPES OF EVALUATION MEASURES OR INDICES**
- ✓ **EXTERNAL OR SUPERVISED INDICES**
 - RAND
 - JACCARD
 - FOWLKES AND MALLOWS
 - Γ STATISTICS

VMail Message	CustServ Calls	Int'l Plan	VMail Plan	Day Charge	Eve Charge	Night Charge	Intl Charge
25	1	0	1	45.07	16.78	11.01	2.7
26	1	0	1	27.47	16.62	11.45	3.7
0	0	0	0	41.38	10.3	7.32	3.29
0	2	1	0	50.9	5.26	8.86	1.78
24	3	0	1	37.09	29.62	9.57	2.03
0	0	1	0	26.69	8.76	9.53	1.92
0	1	0	0	31.37	29.89	9.71	2.35
37	0	1	1	43.96	18.87	14.69	3.02
0	4	0	0	21.95	19.42	9.4	3.43
0	0	0	0	31.91	13.89	8.82	2.46



Your friend, the **CEO OF THE TELECOM COMPANY**, makes the decision to use the **eight attributes** of the above data table to partition the set of customers into **four** different **groups**, each group having its own **phone usage pattern**. This, helps the decision maker to make informed decisions on pricing, special offers, ...

You make the decision to **design and develop a CLUSTER ANALYSIS study** to partition the set of customers of the telecommunication company into four meaningful groups.



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In particular, to plan a **CLUSTER ANALYSIS** study you know that the following issues must be addressed:



- ✓ **SIMILARITY**
- ✓ **EXCLUSIVE OR OVERLAPPING OR FUZZY CLUSTERING**
- ✓ **COMPLETE OR PARTIAL CLUSTERING**



The **CEO OF THE TELECOM COMPANY** thinks that the “ideal number of groups” to partition the set of customers **ranges between 3 and 6**.

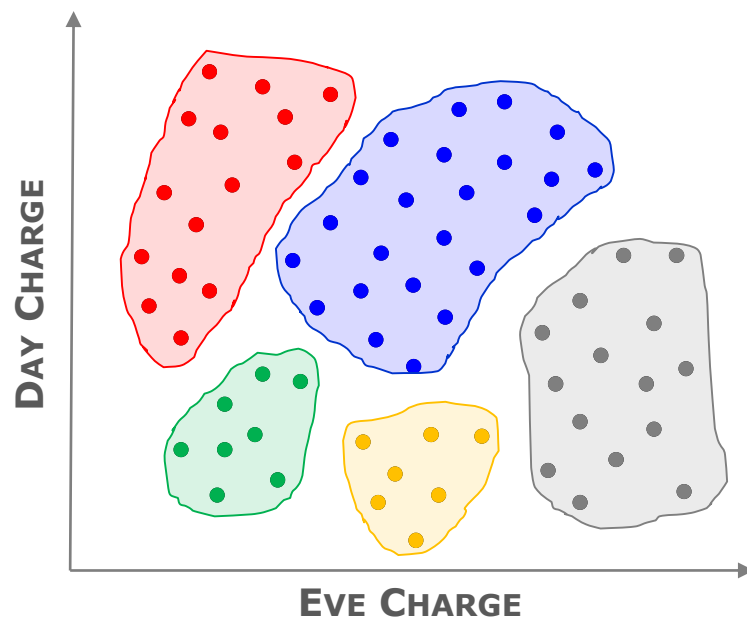
You make the decision to **try different clustering algorithms** with **different parameter values**, i.e. the number of clusters for K-Means and SOMs, or the type of linking (single, average, complete, ...) for agglomerative hierarchical clustering, or *Eps* and *MinPts* for DBSCAN.



You know that in classification (supervised classification), many measures and methods are available to evaluate and compare classification models. Such evaluation measures (accuracy, recall, precision) and methods (cross validation, ...) are well-accepted.

However, because of its very nature, **CLUSTER EVALUATION** is not a **well-developed** or commonly used part of cluster analysis. Nevertheless, cluster evaluation or cluster validation **is of paramount importance**.

To show to the **CEO OF THE TELECOM COMPANY** the difficulty of **CLUSTER ANALYSIS** you use the following example.



You assume that the **two attributes are independent** as it emerges from the plot on the left.

A **clustering algorithm** is **designed to find clusters** and thus it will find clusters in a data set, **even if** that data set has **no natural cluster structure**.

HOW TO EVALUATE THE CLUSTERING RESULT OF A CLUSTERING ALGORITHM?



The most important **ISSUES FOR CLUSTER VALIDATION** are:

- ✓ **DETERMINE THE CLUSTERING TENDENCY** of a set of data, i.e., distinguishing whether non-random structure actually exists in the data
- ✓ **DETERMINE** the "**CORRECT NUMBER OF CLUSTERS**" (whatever it means!!!)

To address the issues of **CLUSTER VALIDATION** we use **EVALUATION MEASURES** or **INDICES**.

EVALUATION MEASURES or **INDICES** are traditionally classified into the following **three types**:

- ✓ **EXTERNAL OR SUPERVISED;** measures the extent to which the clustering structure discovered by a clustering algorithm matches some external structure.
- ✓ **INTERNAL OR UNSUPERVISED;** measures the goodness of a clustering structure without respect to external information. They can be
 - **COHESION MEASURES** (compactness, tightness), which determine how closely related the objects in a cluster are
 - **SEPARATION MEASURES** (isolation), which determine how distinct or well-separated a cluster is from other clusters
- ✓ **RELATIVE;** compares different clusterings or clusters. It can be supervised or unsupervised. Such measures are not a separate type of cluster evaluation measure, but are instead a specific use of such measures.

Let

$$P = \{p_1, \dots, p_R\}$$

be a partitioning, of a data set consisting of " m " objects (records),
into " R " categories.

$$C = \{C_1, \dots, C_K\}$$

be the partition obtained with a clustering algorithm into " K " clusters.

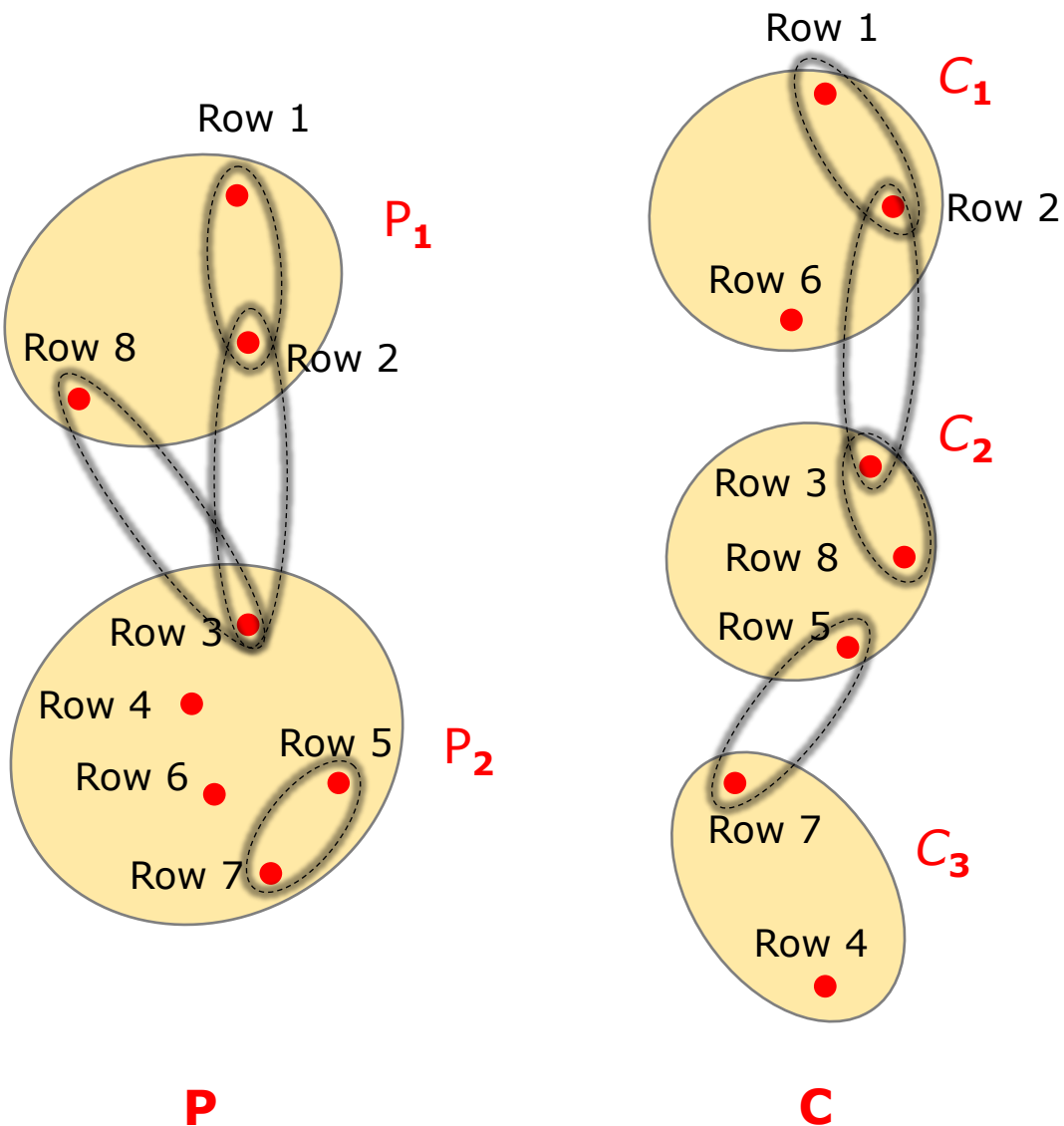
SUPERVISED OR EXTERNAL INDICES compare **P** to **C**:

Case 1: **x** and **y** belong to the same cluster of **C**
and to the same category of **P**

Case 2: **x** and **y** belong to the same cluster of **C**
but to different categories of **P**

Case 3: **x** and **y** belong to different clusters of **C**
but to the same category of **P**

Case 4: **x** and **y** belong to different clusters of **C**
and to different categories of **P**



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SUPERVISED OR EXTERNAL INDICES compare P to C :

- 3 (a) **Case 1:** x and y belong to the same cluster of C and to the same category of P
- 4 (b) **Case 2:** x and y belong to the same cluster of C but to different categories of P
- 10 (c) **Case 3:** x and y belong to different clusters of C but to the same category of P
- 11 (d) **Case 4:** x and y belong to different clusters of C and to different categories of P

The overall number of pairs amounts to

$$M = \frac{m \times (m-1)}{2} = a + b + c + d$$

RAND

$$R = \frac{a + d}{M}$$

$R \in [0,1]$

JACCARD

$$J = \frac{a}{a + b + c}$$

$J \in [0,1]$

**FOWLKES AND
MALLOWS**

$$FM = \sqrt{\frac{a}{a+b} \times \frac{a}{a+c}}$$

$FM \in [0,1]$

Γ STATISTICS

$$\Gamma = \frac{M \times a - (a+b) \times (a+c)}{\sqrt{(a+b) \times (a+c) (M-a-b) \times (M-a-c)}} \quad \Gamma \in [-1,1]$$

The larger the values, the more similar are **C** and **P**.

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SUPERVISED OR EXTERNAL INDICES compare **P** to **C**:

- (a) **Case 1:** **x** and **y** belong to the same cluster of **C** and to the same category of **P**
- (b) **Case 2:** **x** and **y** belong to the same cluster of **C** but to different categories of **P**
- (c) **Case 3:** **x** and **y** belong to different clusters of **C** but to the same category of **P**
- (d) **Case 4:** **x** and **y** belong to different clusters of **C** and to different categories of **P**